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Minimizing Inequity in Facility Location Games

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Abstract

This paper studies the problem of minimizing group-level inequity in facility location games on the real line, where agents belong to different groups and may act strategically. We explore a fairness-oriented objective that minimizes the maximum group effect introduced by Marsh and Schilling (1994). Each group's effect is defined as its total or maximum distance to the nearest facility, weighted by group-specific factors. We show that this formulation generalizes several prominent optimization objectives, including the classical utilitarian (social cost) and egalitarian (maximum cost) objectives, as well as two group-fair objectives, maximum total and average group cost. In order to minimize the maximum group effect, we first propose two novel mechanisms for the single-facility case, the `BALANCED` mechanism and the `MAJOR-PHANTOM` mechanism. Both are strategyproof and achieve tight approximation guarantees under distinct formulations of the maximum group effect objective. Our mechanisms recover the known tight approximation guarantees for classical group-fair objectives as special cases, while extending them to a broader weighted framework and unifying several classical truthful mechanisms.

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1 Introduction

Facility Location Games (FLGs), which study how to locate facilities based on agents' preferences, have been extensively explored over the past two decades. Most prior work in this area has prioritized efficiency, typically aiming to minimize the total cost incurred by agents in accessing services. Such efficiency-driven approaches achieve optimal social welfare, however, at the expense of fairness and equity. In particular, mechanisms designed purely for efficiency tend to favor majority groups, leaving disadvantaged or minority populations marginalized. Recognizing these limitations, recent research has increasingly focused on incorporating fairness into facility location games. These efforts span a spectrum from *individual fairness*, which aims to equalize costs across agents [3, 17], to *group fairness*, which ensures equitable treatment across predefined groups [12, 18, 2]. A seminal contribution by Marsh and Schilling [12] introduced various equity metrics, including the "center" objective, which seeks to **minimize the maximum group effect**. As they note,

> "*This is the earliest and most frequently used measure that has an equity component.*"

This underscores the significance of the center objective in equity-aware location analysis. However, many subsequent studies have adopted a narrow interpretation of this objective, often modeling it as the maximum individual distance across all agents, thereby overlooking group structures and alternative definitions of group-level costs. In contrast, Marsh and Schilling [12] proposed a broader formulation, in which the effect of a group, E_i, could be defined in terms such as the *total distance* incurred by all agents in group i. This generalization more accurately reflects the *collective burden* borne by each group, offering a richer fairness perspective. Motivated by this observation, we revisit the general objective of minimizing $\max_i E_i$ and study its implications within the framework of facility location games.

In this paper, we focus on the objective of minimizing the maximum group effect, where the effect E_i of a group i is defined as either the total or maximum distance from its agents to their nearest facility, multiplied by a weight w_i, capturing group-specific priorities, such as socioeconomic status or policy-driven importance (see Section 2 for formal definitions). Our objective of minimizing $\max_i E_i$ emphasizes group-level fairness by bounding the worst-case burden among all groups. Unlike traditional formulations that protect only the most distant individual, our model accounts for the collective experience of each group. This perspective aligns with Rawlsian principles [16], which advocate prioritizing the welfare of the most disadvantaged. We aim to design new strategyproof mechanisms to this group-centric fairness objective.

1.1 Related Work

FLGs have received significant attention in the literature over the past decades, particularly following the influential work of Procaccia and Tennenholtz [15]. For an overview of the diverse models, we refer readers to the comprehensive survey by Chan et al. [4]. In the remainder of this section, we focus specifically on research that investigates fairness notions within the context of facility location games.

There is a rich body of work studying fairness considerations in facility location problems from the optimization perspective. Early studies in the *operations research* community explored various equity-based fairness measures, including the standard deviation of distances [13] and the Gini coefficient [12]. In the context of algorithmic mechanism design, the pioneering work of Procaccia and Tennenholtz [15] introduced the notion of individual fairness through the maximum cost objective, i.e., minimizing the maximum individual distance from any agent to the facility, and proposed strategyproof mechanisms that approximately optimize this objective. Building on this foundation, later research proposed alternative formulations of individual fairness. Cai et al. [3] studied the minimax envy objective, which captures fairness via the maximum difference in distances between any pair of agents; this framework was subsequently extended to the two-facility setting by Chen et al. [5]. Ding et al. [6] introduced the envy ratio objective, adapted from the fair division problem, which measures the ratio of the best-off agent's utility to that of the worst-off agent. This concept was further extended to the multi-facility case by Liu et al. [11]. Walsh [17] studied the Gini index objective and proposed strategyproof mechanisms.

Fairness notions have also been extended from individuals to groups of agents. Zhou et al. [18], Li et al. [9] investigated two group-based fairness objectives: the *maximum total cost* (mtgc) and *maximum average cost* (magc), each capturing the worst-case burden across predefined groups of agents. Zhou et al. [19] studied these group-fair objectives in a setting with altruistic agents. Aziz et al. [1] extended these objectives to a multi-period mobile facility location setting, where agents from different groups arrive over time and multiple facilities are located in each period; in the single-facility, single-period case, they obtained tight 2-approximation bounds for both objectives, improving the previous approximation guarantee of 3 [18].

Li et al. [10] further incorporated group externalities into this framework, considering both competitive and collaborative interactions among group members. For the competitive setting, they obtained tight 2-approximation bounds for both mtgc and magc. Aziz et al. [2] introduced a model of proportional fairness, in which fairness guarantees are provided to endogenously defined groups of agents, and the strength of the guarantee scales proportionally with group size. This concept was further extended by Lam et al. [8] to the setting of obnoxious facility location, where the facility imposes disutility rather than providing benefit.

Roadmap Section 2 introduces the group-based facility location game model, formally defines our key objective, maximum group effect, and provides an overview of our main contributions. Section 3 focuses on the single-facility setting, where we propose two novel mechanisms tailored to distinct formulations of the maximum group effect objective. In Section 4, we extend our analysis to the multi-facility setting and revisit the classical ENDPOINT mechanism within the framework. Due to space constraints, some proofs are omitted.

2 Model and Contributions

2.1 Facility Location Games

For any $t \in \mathbb{N}$, let $[t] := \{1, 2, \ldots, t\}$. A facility location game consists of a set $N = [n]$ of n agents, belonging to m groups. For each agent $i \in N$, her type is denoted as $\theta_i = (x_i, g_i)$ where $x_i \in \mathbb{R}$ is the agent's *private* location on a line, and $g_i \subseteq [m]$ denotes the set of their *public* group memberships. The type profile of all agents is denoted by $\boldsymbol{\theta} = (\theta_1, \theta_2, \ldots, \theta_n)$. Without loss of generality, we assume that agents are indexed by $x_1 \leq x_2 \leq \cdots \leq x_n$. Let G be the set of groups, $\mathcal{G} = \{G_1, G_2, \ldots, G_m\}$ where $G_j = \{i \in N : j \in g_i\}$ represents the set of agents belonging to group j. Denote by $|G_j|$ the cardinality of G_j. For each group $j \in [m]$, j is assigned with a weight $w_j \geq 0$, reflecting their priority, such as socioeconomic factors or policy-driven importance. To simplify notation, let $w_{\max} = \max_{j\in[m]} w_j$ and $w_{\min} = \min_{j\in[m]} w_j$ denote the maximum and minimum group weights, respectively, and let $w_{g_i} = \max_{g\in g_i}\{w_g\}$ denote the maximum weight among all the groups to which agent i belongs. A deterministic mechanisms $f : \Theta^n \to \mathbb{R}^k$ maps the type profile $\boldsymbol{\theta}$ to locations of k facilities on a real line. Given any mechanism f, for each agent i, the cost incurred by i is defined as $c(f(\boldsymbol{\theta}), x_i) = \min_{y\in f(\boldsymbol{\theta})} |y - x_i|$, i.e., the distance from x_i to the nearest facility.

In this paper, we primarily focus on designing *strategyproof* mechanisms. A mechanism f is said to satisfy **strategyproofness** (SP) if it is in the best interests of every agent i to report their truthful location x_i, irrespectively of the reports of the other agents.

Definition 2.1 (Strategyproofness (SP))**.** *A mechanism f is strategyproof if, for any agent i with true location x_i and group g_i, any misreported location $x_i' \in \mathbb{R}$, and any profile $\boldsymbol{\theta}'_{-i}$ of other agents' reports, we have:*

$$c(f((x_i, g_i), \boldsymbol{\theta}'_{-i}), x_i) \leq c(f((x_i', g_i), \boldsymbol{\theta}'_{-i}), x_i).$$

2.2 Maximum Group Effect

Given the requirement of strategyproofness, our goal is to design mechanisms that minimize inequity, as captured by the maximum group effect objective, aligning with the central notion proposed by Marsh and Schilling [12]. Formally, for any profile $\boldsymbol{\theta}$, the objective is to minimize the **maximum group effect** (mge), defined as

$$\mathsf{mge}(\boldsymbol{\theta}, f(\boldsymbol{\theta})) = \max_{j\in[m]} E_j.$$

We further interpret the *maximum group effect* E_j for each group j in both *utilitarian* and *egalitarian* manners, incorporating the group-specific weight w_j in both formulations. Specifically, one is termed **Weighted Total Group Cost** (wTGC), $E_j = w_j \cdot \sum_{i\in G_j} c(f(\boldsymbol{\theta}), x_i)$, corresponding to the weighted sum of distances from all agents in group G_j to their assigned facilities. The other is termed **Weighted Maximum Group Cost** (wMGC), i.e., $E_j = w_j \cdot \max_{i\in G_j} c(f(\boldsymbol{\theta}), x_i)$, representing the weighted maximum distance among agents in group G_j. mge prioritizes fairness by limiting the worst-case weighted burden among groups, aligning with principles of equitable resource allocation.

For any strategyproof mechanism, we evaluate the performance by the **approximation ratio**, defined as the worst-case ratio (over all possible instances) between the maximum group effect produced by the mechanism and the optimal solution.

Definition 2.2 (Approximation Ratio)**.** *For any mechanism f, the approximation ratio is:*

$$\rho = \sup_{\boldsymbol{\theta} \in \Theta^n} \frac{\mathsf{mge}(\boldsymbol{\theta}, f(\boldsymbol{\theta}))}{\mathsf{mge}(\boldsymbol{\theta}, \mathsf{OPT}(\boldsymbol{\theta}))},$$

where $\mathsf{OPT}(\boldsymbol{\theta})$ *is the optimal facility placement that minimizes* mge *objective under the profile* $\boldsymbol{\theta}$*.*

In the following sections, we focus on deterministic strategyproof mechanisms that optimize the mge objective. We begin with the single-facility setting ($k = 1$) and then extend our analysis to the multi-facility scenario.

2.3 Our Contribution

We advance the field of fair mechanism design in FLGs by introducing a unified framework that seamlessly integrates efficiency, individual fairness, and group fairness.

Capturing Fairness Through Generalized Metrics. We first introduce the general metric termed maximum group effect (mge), which is defined as $\mathsf{mge} = \max_{j \in [m]} E_j$. Here E_j is interpreted either the *weighted total group cost* (wTGC), i.e., $E_j = w_j \cdot \sum_{i \in G_j} c(f(\boldsymbol{\theta}), x_i)$, or the *weighted maximum group cost* (wMGC), i.e., $E_j = w_j \cdot \max_{i \in G_j} c(f(\boldsymbol{\theta}), x_i)$. Our proposed mge objective offers a unified framework for analyzing efficiency and fairness in facility location problems, which captures a broad range of objectives by appropriately adjusting the group partitioning and weight assignments, as illustrated in Figure 1.

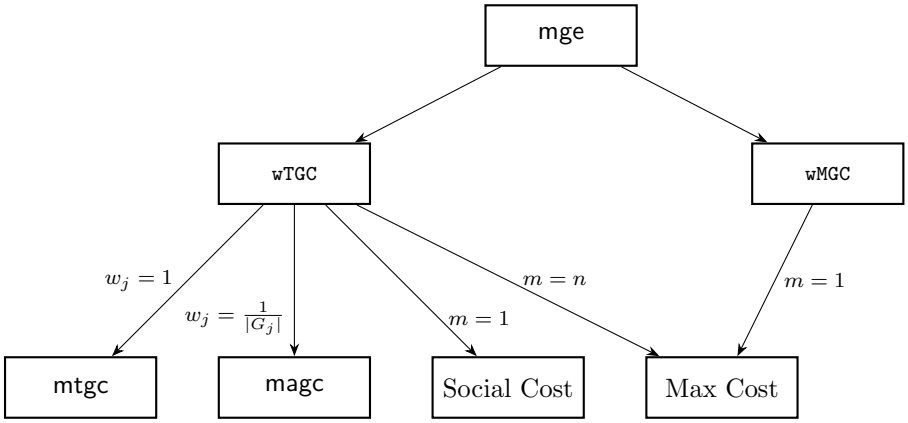


Figure 1. mge: A Systematic Generalization of Objective Metrics.

Unified Mechanisms with Tight Approximation Guarantee. We propose two novel strategyproof mechanisms: the BALANCED mechanism and the MAJOR-PHANTOM mechanism. The BALANCED mechanism, a flagship contribution for single-facility location games, not only unifies classic facility location mechanisms, but also provides tight results for group-fairness objectives. Specifically, regarding the social cost objective, the BALANCED mechanism aligns with the median-point mechanism, while for the maximum cost objective, it degenerates to the LEFTMOST mechanism. In the context of group fairness, the BALANCED mechanism specializes to a 2-approximation for both the maximum total group cost (mtgc) and maximum average group cost (magc), matching the known tight bounds. More generally, we prove that BALANCED achieves a tight 2-approximation for arbitrary weighted total group costs, allowing overlapping group memberships.

Consequently, this unification establishes the BALANCED mechanism as a versatile instrument capable of adapting to diverse fairness and efficiency goals without bespoke designs for objectives. To optimize the weighted maximum group cost objective, we introduce the MAJOR-PHANTOM mechanism and show it provides tight results for this objective.

In the two-facility setting (see Section 4), we revisit the ENDPOINT mechanism [15], which places facilities at the leftmost and rightmost agent locations. For both wTGC and wMGC, we show it achieves tight bounds. For settings beyond two facilities ($k > 2$), we leverage results from Fotakis and Tzamos [7] to show that all strategyproof, anonymous, and deterministic mechanisms yield unbounded approximation ratios. Our established tight approximation ratios and matching lower bounds are comprehensively summarized in Table 1.

Setting	Objectives	Mechanisms	Bounds
$k = 1$	wTGC	BALANCED	2
	wMGC	MAJOR-PHANTOM	2
$k = 2$	wTGC	ENDPOINT	$1 + (n-2)\frac{w_{\max}}{w_{\min}}$
	wMGC	ENDPOINT	$1 + \frac{w_{\max}}{w_{\min}}$
$k \geq 3$	General	/	∞

Table 1. Summary of results. All listed bounds are tight. Gray shading denotes the contributions of this work.

3 Single-Facility Mechanism Design

We begin by considering single-facility setting ($k = 1$). For the wTGC group effect metric, i.e., $E_j = w_j \cdot \sum_{i \in G_j} c(f(\boldsymbol{\theta}), x_i)$, we propose the BALANCED mechanism which places the facility at the location under which the maximum weighted values are balanced. Regarding the wMGC objective where $E_j = w_j \cdot \max_{i \in G_j} c(f(\boldsymbol{\theta}), x_i)$, we introduce the MAJOR-PHANTOM mechanism, which places the facility at the median-point of the locations of agents in the group $G_{\max}$ with the largest weight, together with $(|G_{\max}| - 1)$ constant phantom points. For both cases, we provide tightness results, showing the optimality of the proposed mechanisms.

3.1 Weighted Total Group Cost

We first look at the weighted total group cost (wTGC) objective, where $E_j = w_j \cdot \sum_{i \in G_j} c(f(\boldsymbol{\theta}), x_i)$. As illustrated in Figure 1, the wTGC metric subsumes several well-studied objectives, including the social cost, maximum cost, and group-fairness objectives mtgc and magc. To address this generalized setting, we introduce our first BALANCED mechanism (Mechanism 1), which places the facility at a location that equilibrates the weighted number of agents on either side of it.

Algorithm 1 BALANCED **Mechanism**

Input: Agent profile $\boldsymbol{\theta}$, group weights $\{w_j\}_{j \in [m]}$.

1: Define $L_j(y) \leftarrow |\{i \in N : x_i \leq y \text{ and } j \in g_i\}|$ and $R_j(y) \leftarrow |\{i \in N : x_i > y \text{ and } j \in g_i\}|$.

2: Compute

$$f(\boldsymbol{\theta}) \leftarrow \min\left\{ y \in \mathbb{R} : \max_{j \in [m]} w_j L_j(y) \geq \max_{j' \in [m]} w_{j'} R_{j'}(y) \right\}.$$

Output: Facility location $f(\boldsymbol{\theta})$.

Intuitively, the BALANCED mechanism defines, for each group $j \in [m]$ and location $y \in \mathbb{R}$, two functions $L_j(y)$ and $R_j(y)$, representing respectively the number of agents in group j located at or to the left of y, and those located to its right. Since $w_j L_j(y)$ is non-decreasing and $w_j R_j(y)$ is non-increasing over $y \in [x_1, x_n]$, the mechanism places the facility at a location that most closely balances the two quantities $\max_{j \in [m]} w_j L_j(y)$ and $\max_{j \in [m]} w_j R_j(y)$. The BALANCED mechanism can be implemented in $O((n+m) \log n)$ time by performing a binary search over the sorted agent locations to identify the smallest x_i satisfying $\max_{j \in [m]} w_j L_j(x_i) \geq \max_{j' \in [m]} w_{j'} R_{j'}(x_i)$. Each iteration of the binary search requires evaluating these maximum functions, which takes $O(n + m)$ time.

Proposition 3.1. *The* BALANCED *mechanism coincides with the median-point mechanism (resp. leftmost mechanism) when the* mge *objective degenerates to the social cost (resp. maximum cost) objective.*

We next present the main theorem for the BALANCED mechanism, which satisfies strategyproofness and achieves a 2-approximation for minimizing the mge objective when $E_j = w_j \cdot \sum_{i \in G_j} c(f(\boldsymbol{\theta}), x_i)$ for all $j \in [m]$.

Theorem 3.2. *The* BALANCED *mechanism is strategyproof and has an approximation ratio of* 2 *for minimizing* mge *when* $E_j = w_j \cdot \sum_{i \in G_j} c(f(\boldsymbol{\theta}), x_i)$.

Proof. **Strategyproofness**. Let $f(\boldsymbol{\theta})$ denote the outcome of the BALANCED mechanism, and consider any agent i with truthful location x_i. We prove by discussing the relative positions of $f(\boldsymbol{\theta})$ and x_i. Clearly, if x_i coincides with $f(\boldsymbol{\theta})$, agent i has no incentive to misreport her location. Case (1). If $x_i < f(\boldsymbol{\theta})$, misreporting $x'_i < f(\boldsymbol{\theta})$ will not change the facility location as $L_j(f(\boldsymbol{\theta}))$ and $R_j(f(\boldsymbol{\theta}))$ don't change for all $j \in [m]$. If agent i misreports to $x'_i \geq f(\boldsymbol{\theta})$, we have that $L_j(f(\boldsymbol{\theta}))$ decreases and $R_j(f(\boldsymbol{\theta}))$ increases for each $j \in g_i$, potentially shifting the facility location rightward as $\max_{j' \in [m]}\{w_{j'} R_{j'}(f(\boldsymbol{\theta}))\}$ increases while $\max_{j \in [m]} w_j\{L_j(f(\boldsymbol{\theta}))\}$ decreases. Consequently, agent i's cost increases as the facility moves farther from x_i, implying that misreporting cannot be beneficial. We next consider Case (2). If $x_i > f(\boldsymbol{\theta})$, similarly, when misreporting $x'_i > f(\boldsymbol{\theta})$, the facility location remains at $f(\boldsymbol{\theta})$. When $x'_i \leq f(\boldsymbol{\theta})$, by an analogical induction, it could only potentially pushing the facility location farther away from agent i's location x_i. Hence, we conclude that for any agent $i \in N$, i has no incentive to misreport her location, which implies the BALANCED mechanism is strategyproof.

Approximation Ratio. Denote by $f(\boldsymbol{\theta})$ the BALANCED mechanism outcome and $y^* = \arg\min_{y \in \mathbb{R}} \mathsf{mge}(\boldsymbol{\theta}, y)$ the optimal location for profile $\boldsymbol{\theta}$. We begin with a key observation that underpins the proof of the approximation ratio. Given any profile $\boldsymbol{\theta}$, we construct a modified profile $\boldsymbol{\theta}'$ by relocating all agents whose positions lie between $f(\boldsymbol{\theta})$ and y^* to the point y^*. Under the construction, we first observe that for each agent i who lies between $f(\boldsymbol{\theta})$ and y^*, $c(f(\boldsymbol{\theta}), x_i)$ increases while $c(y^*, x_i)$ decreases, which follows that $\mathsf{mge}(\boldsymbol{\theta}', f(\boldsymbol{\theta})) \geq \mathsf{mge}(\boldsymbol{\theta}, f(\boldsymbol{\theta}))$ and $\mathsf{mge}(\boldsymbol{\theta}', y^*) \leq \mathsf{mge}(\boldsymbol{\theta}, y^*)$. Let $\rho(\boldsymbol{\theta})$ (resp. $\rho(\boldsymbol{\theta}')$) denote the approximation ratio under $\boldsymbol{\theta}$ (resp. $\boldsymbol{\theta}'$). By construction, we have $\rho(\boldsymbol{\theta}) \leq \rho(\boldsymbol{\theta}')$. Henceforth, we focus exclusively on profiles involving such movements. For the sake of clarity, we will abuse the notation $\boldsymbol{\theta}$ to refer to the modified profile.

Case 1: $f(\boldsymbol{\theta}) \leq y^*$. For each group G_j, the group effects under $f(\boldsymbol{\theta})$ and y^* are expressed as $E_j(f(\boldsymbol{\theta})) = w_j \sum_{i \in G_j} |f(\boldsymbol{\theta}) - x_i|$ and $E_j(y^*) = w_j \sum_{i \in G_j} |y^* - x_i|$. Viewing $E_j(y)$ as a function of location y. its derivative can be expressed as $\frac{\mathrm{d}E_j(y)}{\mathrm{d}y} = w_j \cdot (-L_j(y) + R_j(y))$. Consequently, we derive that

$$E_j(f(\boldsymbol{\theta})) - E_j(y^*) = \int_{f(\boldsymbol{\theta})}^{y^*} w_j \cdot (R_j(y) - L_j(y))\mathrm{d}y.$$

Since there is no agent located in the interval $[f(\boldsymbol{\theta}), y^*)$, the derivative is a constant value. We further have

$$E_j(f(\boldsymbol{\theta})) - E_j(y^*) = w_j(R_j(f(\boldsymbol{\theta})) - L_j(f(\boldsymbol{\theta}))) \cdot (y^* - f(\boldsymbol{\theta})).$$

Recall that $\mathsf{mge}(\boldsymbol{\theta}, f(\boldsymbol{\theta})) = \max_{j \in [m]} E_j(f(\boldsymbol{\theta}))$ and $\mathsf{mge}(\boldsymbol{\theta}, y^*) = \max_{j \in [m]} E_j(y^*)$. We then have

$$\begin{aligned}
\mathsf{mge}(\boldsymbol{\theta}, f(\boldsymbol{\theta})) - \mathsf{mge}(\boldsymbol{\theta}, y^*) &\leq \max_{j \in [m]}\{E_j(f(\boldsymbol{\theta})) - E_j(y^*)\} \\
&\leq \max_{j \in [m]}\{w_j(R_j(f(\boldsymbol{\theta})) - L_j(f(\boldsymbol{\theta})))(y^* - f(\boldsymbol{\theta}))\}.
\end{aligned}$$

On the other hand, there are $L_j(f(\boldsymbol{\theta}))$ agents in each group G_j who are at or on the left of $f(\boldsymbol{\theta})$. It follows that

$$\mathsf{mge}(\boldsymbol{\theta}, y^*) \geq \max_{j \in [m]}\{w_j \cdot L_j(f(\boldsymbol{\theta}))\} \cdot (y^* - f(\boldsymbol{\theta})).$$

Moreover, we have $\max_{j \in [m]}\{w_j R_j(f(\boldsymbol{\theta}))\} \leq \max_{j \in [m]}\{w_j L_j(f(\boldsymbol{\theta}))\}$ as $f(\boldsymbol{\theta})$ is the outcome by BALANCED mechanism. With these inequalities in hand, we derive the approximation ratio ρ of the BALANCED mechanism.

$$\begin{aligned}
\rho &= \frac{\mathsf{mge}(\boldsymbol{\theta}, f(\boldsymbol{\theta}))}{\mathsf{mge}(\boldsymbol{\theta}, y^*)} = 1 + \frac{\mathsf{mge}(\boldsymbol{\theta}, f(\boldsymbol{\theta})) - \mathsf{mge}(\boldsymbol{\theta}, y^*)}{\mathsf{mge}(\boldsymbol{\theta}, y^*)} \\
&\leq 1 + \frac{\max_{j \in [m]}\{w_j(R_j(f(\boldsymbol{\theta})) - L_j(f(\boldsymbol{\theta})))\}(y^* - f(\boldsymbol{\theta}))}{\max_{j \in [m]}\{w_j L_j(f(\boldsymbol{\theta}))\}(y^* - f(\boldsymbol{\theta}))} \\
&\leq 1 + \frac{\max_{j \in [m]}\{w_j R_j(f(\boldsymbol{\theta}))\}}{\max_{j \in [m]}\{w_j L_j(f(\boldsymbol{\theta}))\}} \leq 2.
\end{aligned}$$

Case 2: When $f(\boldsymbol{\theta}) > y^*$. Keep in mind that we still have the derivative expression for function $E_j(y)$. Observe that in the interval $[y^*, f(\boldsymbol{\theta}))$, the derivative of function $E_j(y)$ is a constant value as there is no agent in the interval. By an analogous approach, we have

$$E_j(f(\boldsymbol{\theta})) - E_j(y^*) = w_j(R_j(y^*) - L_j(y^*)) \cdot (f(\boldsymbol{\theta}) - y^*).$$

We next establish the group effect difference between solution $f(\boldsymbol{\theta})$ and y^*.

$$\begin{aligned}\mathsf{mge}(\boldsymbol{\theta}, f(\boldsymbol{\theta})) - \mathsf{mge}(\boldsymbol{\theta}, y^*) &\leq \max_{j\in[m]}\{E_j(f(\boldsymbol{\theta})) - E_j(y^*)\} \\ &\leq \max_{j\in[m]}\{w_j(R_j(y^*) - L_j(y^*))(y^* - f(\boldsymbol{\theta}))\}.\end{aligned}$$

Since we know that there is no agent in the interval $(y^*, f(\boldsymbol{\theta}))$, then for all the agents on the right of y^*, they must satisfy $x_i \geq f(\boldsymbol{\theta})$. Therefore, we can bound $\mathsf{mge}(\boldsymbol{\theta}, y^*)$ by

$$\mathsf{mge}(\boldsymbol{\theta}, y^*) \geq \max_{j\in[m]}\{w_j \cdot R_j(y^*)\} \cdot (f(\boldsymbol{\theta}) - y^*).$$

Based on the aforementioned analysis, we derive the approximation ratio

$$\begin{aligned}\rho &= \frac{\mathsf{mge}(\boldsymbol{\theta}, f(\boldsymbol{\theta}))}{\mathsf{mge}(\boldsymbol{\theta}, y^*)} \\ &\leq 1 + \frac{\max_{j\in[m]}\{w_j(R_j(y^*) - L_j(y^*))\}(f(\boldsymbol{\theta}) - y^*)}{\max_{j\in[m]}\{w_j \cdot R_j(y^*)\}(f(\boldsymbol{\theta}) - y^*)} \\ &\leq 1 + \frac{\max_{j\in[m]}\{w_j R_j(y^*)\}}{\max_{j\in[m]}\{w_j R_j(y^*)\}} = 2.\end{aligned}$$

Combining the analyses of both cases, we conclude that the BALANCED mechanism achieves an approximation ratio of 2 for minimizing the maximum group effect when $E_j = w_j \cdot \sum_{i\in G_j} c(f(\boldsymbol{\theta}), x_i)$. □

Since mge generalizes the maximum cost objective when $m = n$ and $G_j = \{j\}$ with equal weights, the lower bound from Procaccia and Tennenholtz [15] applies, confirming the tightness of BALANCED 's approximation ratio.

Corollary 3.3 (15)**.** *Any deterministic mechanism has an approximation ratio of at least* 2 *for* mge *when* $E_j = w_j \cdot \sum_{i\in G_j} c(f(\boldsymbol{\theta}), x_i)$.

Note that the mge objective also generalizes the group-fair objectives mtgc and magc proposed by Zhou et al. [18]. While Zhou et al. [18] established a 3-approximation upper bound and a lower bound of 2. Subsequent work obtained tight 2-approximation guarantees for these objectives in related extensions [1, 10]. Theorem 3.2 recovers this tight factor as a special case of our more general weighted formulation.

Corollary 3.4. *The* BALANCED *mechanism achieves a* 2*-approximation ratio w.r.t. the* mtgc *and* magc *objectives.*

3.2 Weighted Maximum Group Cost

We next turn to the weighted maximum group cost objective, wherein $E_j = w_j \cdot \max_{i\in G_j} c(f(\boldsymbol{\theta}), x_i)$. Intuitively, it is a weighted maximum cost problem where each agent i is assigned with a maximum weight $w_{g_i} = \max_{j\in g_i} w_j$, and the objective is to minimize the maximum value of $w_{g_i} \cdot c(f(\boldsymbol{\theta}), x_i)$ over all agents $i \in N$. In view of this, we propose the MAJOR-PHANTOM mechanism (Mechanism 2) which selects the facility location by prioritizing the group with the largest weight w_j.

Algorithm 2 MAJOR-PHANTOM **Mechanism**

Input: Agent profile $\boldsymbol{\theta}$, group weights $\{w_j\}_{j\in[m]}$.

1: Let $G_{\max}$ denote the largest weight group and $\mathbf{x}^{G_{\max}} = \{x_1^{G_{\max}}, \ldots, x_{|G_{\max}|}^{G_{\max}}\}$ denote the location profile of agents in G^*, tie-breaking in favor of the smallest index.

2: Let $v_1 \leq \cdots \leq v_{|G_{\max}|-1}$ denote $|G_{\max}| - 1$ values $v_1 \leq \cdots \leq v_{|G_{\max}|-1}$.

3: $f(\boldsymbol{\theta}) \leftarrow \mathrm{med}(\mathbf{x}^{G_{\max}}, v_1, \ldots, v_{|G_{\max}|-1})$, tie-breaking by selecting the leftmost.

Output: Facility location $f(\boldsymbol{\theta})$.

MAJOR-PHANTOM mechanism extends the PHANTOM mechanisms [14] by prioritizing agents in the largest-weighted group, thereby ensuring fairness for groups with greater importance. Before analyzing the approximation ratio of the MAJOR-PHANTOM mechanism with respect to the wMGC objective, we first provide a characterization of the optimal solution in two-agent instances, which will facilitate the subsequent analysis.

Lemma 3.5. *Given two-agent profile $\boldsymbol{\theta}$ and optimal solution y^*, for any two agents with locations $x_1 \le x_2$ and maximum weights $w_{g_1}, w_{g_2} \ge 0$, $\mathsf{mge}(\boldsymbol{\theta}, y^*) = \max_{j\in[m]} w_j \cdot \max_{i\in G_j} |y^* - x_i|$ is either*

- $w_{g_1} \cdot \frac{(x_2-x_1)}{2}$ *when $g_1 = g_2$ with the maximum weight w_{g_1}, achieved at $y^* = \frac{x_1+x_2}{2}$; or*
- $\frac{w_{g_1}\cdot w_{g_2}\cdot(x_2-x_1)}{w_{g_1}+w_{g_2}}$ *when $w_{g_1} \neq w_{g_2}$, achieved at $y^* = \frac{w_{g_2}x_2+w_{g_1}x_1}{w_{g_1}+w_{g_2}}$.*

We next prove that for any MAJOR-PHANTOM mechanism, it is strategyproof and achieves an approximation ratio of 2 for minimizing mge under the wMGC objective.

Theorem 3.6. *Any* MAJOR-PHANTOM *mechanism is strategyproof and has an approximation ratio of 2 for minimizing the* mge *objective when $E_j = w_j \cdot \max_{i\in G_j} c(f(\boldsymbol{\theta}), x_i)$.*

Proof. **Strategyproofness**. Given any agent profile $\boldsymbol{\theta}$, consider an agent $i \in N$ with true location x_i. If $i \notin G_{\max}$, it is clear that misreporting cannot influence the facility location under the mechanism. If $i \in G_{\max}$, then since group membership cannot be misreported, we can apply a similar analytical approach to that used in the proof of strategyproofness for PHANTOM mechanisms by Moulin [14].

Approximation Ratio. Given any profile $\boldsymbol{\theta}$, let $f(\boldsymbol{\theta})$ denote the location outputted by the MAJOR-PHANTOM mechanism and y^* denote the optimal location under $\boldsymbol{\theta}$.

We first consider the case that $y^* \ge f(\boldsymbol{\theta})$. Suppose that $\mathsf{mge}(\boldsymbol{\theta}, f(\boldsymbol{\theta}))$ is achieved by agent ℓ. We first observe that if $x_\ell \le f(\boldsymbol{\theta})$, we have $\mathsf{mge}(\boldsymbol{\theta}, f(\boldsymbol{\theta})) = w_{g_\ell} \cdot (f(\boldsymbol{\theta}) - x_\ell)$ and $\mathsf{mge}(\boldsymbol{\theta}, y^*) \ge w_{g_\ell} \cdot (y^* - x_\ell)$, which gives us

$$\rho = \frac{\mathsf{mge}(\boldsymbol{\theta}, f(\boldsymbol{\theta}))}{\mathsf{mge}(\boldsymbol{\theta}, y^*)} \le \frac{w_{g_\ell} \cdot (f(\boldsymbol{\theta}) - x_\ell)}{w_{g_\ell} \cdot (y^* - x_\ell)} \le 1.$$

If $x_\ell > f(\boldsymbol{\theta})$, let $k \in G_{\max}$ be the agent in group $G_{\max}$ whose location x_k satisfies $x_k = \min_{j\in G_{\max}} |x_j - f(\boldsymbol{\theta})|$ with the additional condition that $x_k \le f(\boldsymbol{\theta})$. That is, x_k is the closest agent to the left of $f(\boldsymbol{\theta})$ within $G_{\max}$[1]. We claim that such an agent k always exists. Toward this end, suppose, for the sake of contradiction, that no such x_k exists. It implies that all the agents' locations $\{x_1^{G_{\max}}, \ldots, x_{|G_{\max}|}^{G_{\max}}\}$ lie strictly to the right of $f(\boldsymbol{\theta})$, i.e., $x_i^{G_{\max}} > f(\boldsymbol{\theta})$ for all $i \in G_{\max}$. However, under the MAJOR-PHANTOM mechanism, the facility is placed at the median of the multiset $\{\mathbf{x}^{G_{\max}}, v_1, \ldots, v_{|G_{\max}|-1}\}$, which has a size of $2\cdot|G_{\max}|-1$. In this case, there can be at most $|G_{\max}|-1$ points strictly to the right of $f(\boldsymbol{\theta})$, contradicting the assumption. Therefore, such an agent k always exists.

If $x_k \le f(\boldsymbol{\theta}) < x_\ell < y^*$, we have $\mathsf{mge}(\boldsymbol{\theta}, f(\boldsymbol{\theta})) = w_{g_\ell} \cdot (x_\ell - f(\boldsymbol{\theta})) \le w_{g_\ell} \cdot (x_\ell - x_k)$ and $\mathsf{mge}(\boldsymbol{\theta}, y^*) \ge w_{g_k} \cdot (y^* - x_k) \ge w_{g_k} \cdot (x_\ell - x_k)$. Hence, the approximation ratio is expressed as

$$\rho = \frac{\mathsf{mge}(\boldsymbol{\theta}, f(\boldsymbol{\theta}))}{\mathsf{mge}(\boldsymbol{\theta}, y^*)} \le \frac{w_{g_\ell} \cdot (x_\ell - x_k)}{w_{g_k} \cdot (x_\ell - x_k)} = \frac{w_{g_\ell}}{w_{g_k}}.$$

Recall the definition of MAJOR-PHANTOM mechanism. We know that $\rho \le \frac{w_{g_\ell}}{w_{g_k}} = \frac{w_{g_\ell}}{w_{\max}} \le 1$ as $w_{g_k} = w_{\max} \ge w_{g_\ell}$.

If $x_k \le f(\boldsymbol{\theta}) < y^* < x_\ell$, we have $\mathsf{mge}(\boldsymbol{\theta}, f(\boldsymbol{\theta})) = w_{g_\ell} \cdot (x_\ell - f(\boldsymbol{\theta})) \le w_{g_\ell} \cdot (x_\ell - x_k)$. By Theorem 3.5, when only considering agent k and ℓ, we have the maximum cost achieved by these two agents is at least $\frac{w_{g_\ell} w_{g_k}(x_\ell - x_k)}{w_{g_\ell}+w_{g_k}}$. Hence, we have $\mathsf{mge}(\boldsymbol{\theta}, y^*) \ge \frac{w_{g_\ell} w_{g_k}(x_\ell - x_k)}{w_{g_\ell}+w_{g_k}}$. Consequently, the approximation ratio is bounded by

$$\rho = \frac{\mathsf{mge}(\boldsymbol{\theta}, f(\boldsymbol{\theta}))}{\mathsf{mge}(\boldsymbol{\theta}, y^*)} \le \frac{w_{g_\ell} \cdot (x_\ell - x_k)}{\frac{w_{g_\ell} w_{g_k}(x_\ell - x_k)}{w_{g_\ell}+w_{g_k}}} = \frac{w_{g_\ell} + w_{g_k}}{w_{g_k}}.$$

Since $w_{g_k} = w_{\max} \ge w_{g_\ell}$, it follows that $\rho = \frac{w_{g_\ell}+w_{g_k}}{w_{g_k}} = \frac{w_{g_\ell}+w_{\max}}{w_{\max}} \le 2$. For the case where $y^* < f(\boldsymbol{\theta})$, the same approximation ratio of 2 can be established by applying an analogous analysis to that used for the case $f(\boldsymbol{\theta}) \ge y^*$. □

Notice that the mge objective coincides with the maximum cost objective when $m = n$ and $G_j = \{j\}$ with equal weights. In this case, the lower bound of 2 for the maximum cost objective established by Procaccia and Tennenholtz [15] applies, thereby confirming the tightness of the bounds achieved by the MAJOR-PHANTOM mechanism.

Corollary 3.7 (15)**.** *Any deterministic, strategyproof mechanism has an approximation ratio of at least 2 for* mge *when $E_j = w_j \cdot \max_{i\in G_j} c(f(\boldsymbol{\theta}), x_i)$.*

[1] If multiple agents satisfy this condition, we break ties by selecting the agent with the largest index k.

4 Multi-Facility Mechanism Analysis

In this section, we extend our analysis from single-facility to multi-facility settings. In view of the impossibility result of Fotakis and Tzamos [7], which shows that for $k \geq 3$, no deterministic, anonymous, and strategyproof mechanism can achieve a bounded approximation ratio for either the social cost or maximum cost objectives, our primary focus is on the two-facility case ($k = 2$).

Corollary 4.1. *When $k \geq 3$, there is no deterministic, anonymous, strategyproof mechanisms with a bounded approximation ratio for* mge*, for either $E_j = w_j \cdot \sum_{i \in G_j} c(f(\boldsymbol{\theta}), x_i)$ or $E_j = w_j \cdot \max_{i \in G_j} c(f(\boldsymbol{\theta}), x_i)$.*

We next restrict our attention to the case of $k = 2$ and revisit the ENDPOINT mechanism (placing facilities at the leftmost and rightmost agent locations), which remains the only known deterministic, anonymous, and strategyproof mechanism with bounded approximation guarantees for these objectives [7].

While Fotakis and Tzamos [7] established that the ENDPOINT mechanism is the only deterministic, anonymous, and strategyproof mechanism with bounded approximation guarantees for social cost in the two-facility setting ($k = 2$), evaluating its performance under our group-centric mge objective presents a novel and non-trivial challenge. Unlike classical objectives, the mge objective requires accounting for weighted group effects, where both the group structures and the distribution of group weights play a critical role, which demand a fundamentally different analytical approach. Our contribution lies in establishing tight approximation bounds for the ENDPOINT mechanism under mge, thereby extending its applicability to equitable facility placement and offering theoretical insights into group fairness.

4.1 Weighted Total Group Cost

We first explore the maximum group effect objective by considering the weighted total group cost (wTGC). Our result shows that the ENDPOINT mechanism achieves an approximation ratio of $1 + (n-2) \cdot \frac{w_{\max}}{w_{\min}}$.

Theorem 4.2. *The* ENDPOINT *mechanism has an approximation ratio of $1 + (n-2)\frac{w_{\max}}{w_{\min}}$ for minimizing* mge *when $E_j = w_j \cdot \sum_{i \in G_j} c(f(\boldsymbol{\theta}), x_i)$.*

Proof. Given any agent profile $\boldsymbol{\theta}$, let $Y = (x_1, x_n)$ denote the outputs of the ENDPOINT mechanism and $Y^* = (y_1^*, y_2^*)$ (w.l.o.g, $y_1^* \leq y_2^*$) denote the optimal facility locations which achieves optimal $\mathsf{mge}(\boldsymbol{\theta}, Y^*)$. We first observe that $x_1 \leq y_1^* \leq y_2^* \leq x_n$. For any group G_j, suppose there are k_1^j agents (excluding agent 1) who are assigned to facility y_1^* while k_2^j agents (excluding agent n) assigned to facility y_2^*[2]. Now we consider the follow movement, moving one facility from y_1^* to x_1 and the other facility from y_2^* to x_n. For group G_j, after the movement, the changes of the group effect is expressed as

$$\begin{aligned} E_j(Y) - E_j(Y^*) &\leq w_j \left(k_1^j (y_1^* - x_1) + k_2^j (x_n - y_2^*) \right) \\ &\leq w_j (k_1^j + k_2^j) \cdot \max\{y_1^* - x_1, x_n - y_2^*\} \\ &\leq w_j (n-2) \cdot \max\{y_1^* - x_1, x_n - y_2^*\}. \end{aligned}$$

Recall that $w_{\max} = \max_j w_j$ and $w_{\min} = \min_j w_j$. Now we consider the mge objective and have

$$\begin{aligned} \mathsf{mge}(\boldsymbol{\theta}, Y) - \mathsf{mge}(\boldsymbol{\theta}, Y^*) &\leq \max_{j \in [m]} (E_j(Y) - E_j(Y^*)) \\ &\leq w_{\max} \cdot (n-2) \cdot \max\{y_1^* - x_1, x_n - y_2^*\}. \end{aligned} \tag{1}$$

On the other hand, since there exists at least one agent who is assigned to each facility under the optimal solution Y^*, we have the lower bound that

$$\begin{aligned} \mathsf{mge}(\boldsymbol{\theta}, Y^*) &\geq \max\{w_{g_1}(y_1^* - x_1), w_{g_n}(x_n - y_2^*)\} \\ &\geq w_{\min} \cdot \max\{y_1^* - x_1, x_n - y_2^*\}. \end{aligned} \tag{2}$$

[2]Breaking ties by assigning to y_1^*

From Equation (1) and Equation (2), we derive the upper bound of the approximation ratio ρ

$$\begin{aligned}\frac{\mathsf{mge}(\boldsymbol{\theta}, Y)}{\mathsf{mge}(\boldsymbol{\theta}, Y^*)} \leq & 1 + \frac{(n-2)w_{\max} \cdot \max\{y_1^* - x_1, x_n - y_2^*\}}{\max\{w_{g_1}(y_1^* - x_1), w_{g_n}(x_n - y_2^*)\}} \\ \leq & 1 + \frac{(n-2)w_{\max} \cdot \max\{y_1^* - x_1, x_n - y_2^*\}}{w_{\min} \max\{y_1^* - x_1, x_n - y_2^*\}} \\ \leq & 1 + (n-2)\frac{w_{\max}}{w_{\min}}.\end{aligned}$$

To show the tightness, consider an instance with n agents where $x_1 = 0, x_2 = x_3 = \cdots = x_{n-1} = \frac{1}{2}$, and $x_n = 1$, and $G_1 = \{1\}, G_2 = \{2, 3, \ldots, n\}$. For group weights, let $w_1 = w_{\min}$ and $w_2 = w_{\max}$. We first observe the optimal solution $Y^* = (y_1^* = \frac{w_{\max}(n-2)}{2[w_{\min}+w_{\max}(n-2)]}, y_2^* = 1)$, achieving $\mathsf{mge}(\boldsymbol{\theta}, Y^*) = \frac{w_{\min}w_{\max}(n-2)}{2[w_{\min}+w_{\max}(n-2)]}$. In contrast, the ENDPOINT mechanism has an mge of $\frac{w_{\max}(n-2)}{2}$. This gives us an approximation ratio of $1 + (n-2) \cdot \frac{w_{\max}}{w_{\min}}$. □

We adapt the characterization of Fotakis and Tzamos [7], which identifies the ENDPOINT mechanism as the unique deterministic, anonymous, and strategyproof mechanism with bounded approximation ratio for $k = 2$, to establish the tightness.

Proposition 4.3. *Any deterministic, strategyproof mechanism has an approximation ratio of at least* $1 + (n-2) \cdot \frac{w_{\max}}{w_{\min}}$ *when* $E_j = w_j \cdot \sum_{i \in G_j} c(f(\boldsymbol{\theta}), x_i)$.

4.2 Weighted Maximum Group Cost

We now turn to the weighted maximum group cost (wMGC) objective. Under this criterion, the ENDPOINT mechanism attains an approximation ratio of $(1 + \frac{w_{\max}}{w_{\min}})$.

Theorem 4.4. *The* ENDPOINT *mechanism has an approximation ratio of* $1 + \frac{w_{\max}}{w_{\min}}$ *for minimizing* mge *when* $E_j = w_j \cdot \max_{i \in G_j} c(f(\boldsymbol{\theta}), x_i)$.

Proof. Given any agent profile $\boldsymbol{\theta}$, Denote by $Y = (x_1, x_n)$ the outputs of the ENDPOINT mechanism and $Y^* = (y_1^*, y_2^*)$ $(y_1^* \leq y_2^*)$ the optimal facility placement which achieves optimal $\mathsf{mge}(\boldsymbol{\theta}, Y^*)$. We first observe that $x_1 \leq y_1^* \leq y_2^* \leq x_n$. Without loss of generality, assume $\mathsf{mge}(\boldsymbol{\theta}, Y)$ is achieved by agent k and $x_k \leq \frac{x_1+x_n}{2}$, i.e., k is assigned to facility located at x_1 under Y.

Case 1. When $x_1 \leq x_k \leq y_1^*$, we have $\mathsf{mge}(\boldsymbol{\theta}, Y) = w_{g_k}(x_k - x_1)$ and $\mathsf{mge}(\boldsymbol{\theta}, Y^*) \geq w_{g_1}(y_1^* - x_1)$. The approximation ratio ρ is upper-bounded by

$$\frac{\mathsf{mge}(\boldsymbol{\theta}, Y)}{\mathsf{mge}(\boldsymbol{\theta}, Y^*)} \leq \frac{w_{g_k} \cdot (x_k - x_1)}{w_{g_1} \cdot (y_1^* - x_1)} \leq \frac{w_{g_k} \cdot (x_k - x_1)}{w_{g_1} \cdot (x_k - x_1)} \leq \frac{w_{\max}}{w_{\min}}.$$

Case 2. When $y_1^* < x_k < y_2^*$ and k is assigned to y_1^*. By Theorem 3.5, we have $\mathsf{mge}(\boldsymbol{\theta}, Y^*) \geq \frac{w_{g_1} w_{g_k}(x_k - x_1)}{w_{g_1} + w_{g_k}}$ and the approximation ratio ρ is upper-bounded by

$$\frac{\mathsf{mge}(\boldsymbol{\theta}, Y)}{\mathsf{mge}(\boldsymbol{\theta}, Y^*)} \leq \frac{w_{g_k} \cdot (x_k - x_1)}{\frac{w_{g_1} w_{g_k}(x_k - x_1)}{w_{g_1} + w_{g_k}}} \leq 1 + \frac{w_{g_k}}{w_{g_1}} \leq 1 + \frac{w_{\max}}{w_{\min}}.$$

Case 3. When $y_1^* < x_k < y_2^*$ and agent k is assigned to y_2^*. Similarly, we can lower-bound $\mathsf{mge}(\boldsymbol{\theta}, Y^*) \geq \frac{w_{g_k} w_{g_n}(x_n - x_k)}{w_{g_k} + w_{g_n}}$. Recall that $x_k \leq \frac{x_1+x_n}{2}$. It implies that $x_k - x_1 \leq x_n - x_k$. So we have the lower bound for the approximation ratio ρ that

$$\frac{\mathsf{mge}(\boldsymbol{\theta}, Y)}{\mathsf{mge}(\boldsymbol{\theta}, Y^*)} \leq \frac{w_{g_k} \cdot (x_k - x_1)}{\frac{w_{g_k} w_{g_n}(x_n - x_k)}{w_{g_k} + w_{g_n}}} \leq 1 + \frac{w_{g_k}}{w_{g_n}} \leq 1 + \frac{w_{\max}}{w_{\min}}.$$

Case 4. When $y_2^* \leq x_k \leq x_n$. In this case, agent k is assigned to the facility located at y_2^*. Hence we derive that $\mathsf{mge}(\boldsymbol{\theta}, Y^*) \geq w_n(x_n - y_2^*)$. Consequently, the approximation ratio ρ satisfies $\rho = \frac{\mathsf{mge}(\boldsymbol{\theta}, Y)}{\mathsf{mge}(\boldsymbol{\theta}, Y^*)} \leq$

$\frac{w_{g_k}\cdot(x_k-x_1)}{w_n\cdot(x_n-y_2^*)}$. Note that we also have $x_k - x_1 \leq x_n - x_k$ and $x_n - y_2^* \geq x_n - x_k$. Therefore, we bound the approximation ratio ρ by

$$\frac{\mathsf{mge}(\boldsymbol{\theta}, Y)}{\mathsf{mge}(\boldsymbol{\theta}, Y^*)} \leq \frac{w_{g_k}\cdot(x_k - x_1)}{w_n\cdot(x_n - y_2^*)} \leq \frac{w_{g_k}\cdot(x_n - x_k)}{w_n\cdot(x_n - x_k)} \leq \frac{w_{\max}}{w_{\min}}.$$

Combining the analysis across all four cases, we conclude that the `ENDPOINT` mechanism achieves an approximation ratio of $1 + \frac{w_{\max}}{w_{\min}}$. To establish the tightness of this bound, consider the following instance. There are n agents where $x_1 = x_2 = \cdots = x_{n-2} = 0$, $x_{n-1} = \frac{1}{2}$, and $x_n = 1$. The group structure is given by $G_1 = \{1\}$, and $G_2 = \{2, 3, \ldots, n\}$ with weights $w_1 = w_{\min}$, and $w_2 = w_{\max}$. We first identify that the optimal solution is $Y^* = (y_1^* = \frac{w_{\max}}{2(w_{\max}+w_{\min})}, y_2^* = 1)$, achieving an mge value of $\frac{w_{\min}\cdot w_{\max}}{2(w_{\min}+w_{\max})}$. In contrast, the `ENDPOINT` mechanism attains an mge value of $\frac{w_{\max}}{2}$, implying that it has an approximation ratio of $1 + \frac{w_{\max}}{w_{\min}}$. □

Proposition 4.5. *Any deterministic, strategyproof mechanism has an approximation ratio of at least* $1+\frac{w_{\max}}{w_{\min}}$ *when* $E_j = w_j \cdot \max_{i\in G_j} c(f(\boldsymbol{\theta}), x_i)$.

5 Conclusion and Discussion

We study facility location games through the lens of fairness by introducing a unified framework based on the *maximum group effect*, a general metric that encompasses a broad class of classical objectives. In the single-facility setting, we develop two strategyproof mechanisms, `BALANCED` and `MAJOR-PHANTOM`, both of which achieve tight approximation guarantees for minimizing the maximum group effect. Our results recover the known tight bounds for the classical group-fair objectives as special cases and extend these guarantees to weighted and overlapping group structures. In the two-facility setting, we revisit the classical `ENDPOINT` mechanism and establish tight approximation bounds. Looking forward, promising research directions include extending our framework to randomized mechanisms to circumvent the impossibility of achieving bounded approximations for $k \geq 3$, as well as adapting the mge objective to higher dimensional metric spaces.

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A Comparative Analysis

In this section, we revisit three classical mechanisms introduced by Procaccia and Tennenholtz [15] and Zhou et al. [18]: the MEDIAN mechanism, the LEFTMOST mechanism, and the Majority Group Median (MAJOR) Mechanism. We then analyze the approximation ratios of these mechanisms under the maximum group effect objective, showing that our proposed mechanisms, BALANCED and MAJOR-PHANTOM, achieve superior performance in approximating optimum for both the wTGC and wMGC objectives. A summary of the comparative results is presented in Table 2.

Objectives	Mechanisms	Bounds
wTGC	MEDIAN	$1+(m-1)\cdot\frac{w_{\max}}{w_{\min}}$
	LEFTMOST	$1+(n-1)\cdot\frac{w_{\max}}{w_{\min}}$
	MAJOR	$1+2\cdot\frac{w_{\max}}{w_{\min}}$
	BALANCED	**2**
wMGC	MEDIAN LEFTMOST MAJOR	$1+\frac{w_{\max}}{w_{\min}}$
	MAJOR-PHANTOM	**2**

Table 2. Summary of the comparative analysis

The MEDIAN mechanism and LEFTMOST mechanism are the fundamental mechanisms studied by Procaccia and Tennenholtz [15]. Given any agent profile $\boldsymbol{\theta}$, without loss of generality, we assume that $x_1 \leq x_2 \leq \cdots \leq x_n$. The MEDIAN mechanism places the facility at $f(\boldsymbol{\theta}) = x_{\lceil \frac{n}{2} \rceil}$ and the LEFTMOST mechanism places the facility at $f(\boldsymbol{\theta}) = x_1$. Regarding the Majority Group Median (MAJOR) Mechanism by Zhou et al. [18]. It places the facility at the median point among the largest-size group agents.

Algorithm 3 MAJOR **Mechanism**

Input: Agent profile $\boldsymbol{\theta}$, group weights $\{w_j\}_{j\in[m]}$.

1: Let G_ℓ denote the largest size group and $\mathbf{x}_\ell = \{x^\ell_1, \ldots, x^\ell_{|G_\ell|}\}$ be the location profile of agents, tie-breaking in favor of the smallest index.

2: Locate the facility $f(\boldsymbol{\theta})$ at the median point of multiset $\mathbf{x} = \{x^\ell_1, \ldots, x^\ell_{|G_\ell|}\}$, tie-breaking by select the leftmost.

Output: Facility location $f(\boldsymbol{\theta})$.

We next study the approximation performance for these three mechanisms in terms of both the weighted total group cost (wTGC) objective and the weighted maximum group cost (wMGC) objective when placing one single facility ($k = 1$).

A.1 Weighted Total Group Cost

We begin by considering the weighted group cost (wTGC) objective, defined as $E_j = w_j \cdot \sum_{i\in G_j} c(f(\theta), x_i)$. We slightly abuse notation by using the functions $L_j(y)$ and $R_j(y)$ to denote the number of agents located to the left of any location y and to the right of y for any group $G_j \in \mathcal{G}$, respectively. Throughout, let $w_{\max}$ and $w_{\min}$ denotes the maximum weight and the minimum group weights.

Proposition A.1. *The* MEDIAN *mechanism has an approximation ratio of* $1+(m-1)\cdot\frac{w_{\max}}{w_{\min}}$ *for minimizing* mge *when* $E_j = w_j \cdot \sum_{i\in G_j} c(f(\boldsymbol{\theta}), x_i)$.

Proof. Consider an arbitrary profile $\boldsymbol{\theta}$. Let $f(\boldsymbol{\theta})$ be the output of the MEDIAN mechanism and y^* be the optimal facility location. Without loss of generality, assume that $f(\boldsymbol{\theta}) < y^*$ and that $\mathsf{mge}(\boldsymbol{\theta}, f(\boldsymbol{\theta}))$ is attained by some group G_g. We first claim that, the value of $\mathsf{mge}(\boldsymbol{\theta}, f(\boldsymbol{\theta}))$ is non-decreasing, while $\mathsf{mge}(\boldsymbol{\theta}, y^*)$ is non-increasing when all the agents located between $f(\boldsymbol{\theta})$ and y^* are relocated to y^*. Consequently, the approximation ratio of $f(\boldsymbol{\theta})$ under $\boldsymbol{\theta}$ is upper-bounded by that of the resulting instance after the relocation. Henceforth, we restrict attention to instances in which no agent is located strictly between $f(\boldsymbol{\theta})$ and y^*.

Note that $\mathsf{mge}(\boldsymbol{\theta}, f(\boldsymbol{\theta})) = w_g \sum_{i \in G_g} d(x_i, f(\boldsymbol{\theta}))$ and by the definition $\mathsf{mge}(\boldsymbol{\theta}, y^*) \geq w_g \sum_{i \in G_g} d(x_i, y^*)$. We derive the difference between the two terms by

$$\begin{aligned}\mathsf{mge}(\boldsymbol{\theta}, f(\boldsymbol{\theta})) - \mathsf{mge}(\boldsymbol{\theta}, y^*) &\leq w_g \sum_{i \in G_g} d(x_i, f(\boldsymbol{\theta})) - w_g \sum_{i \in G_g} d(x_i, y^*) \\ &\leq w_g(|R_g(f(\boldsymbol{\theta}))| - |L_g(f(\boldsymbol{\theta}))|)(y^* - f(\boldsymbol{\theta})).\end{aligned}$$

Observe that the mge objective is at least the sum of the costs of agents in the same group that are positioned to the left of $f(\boldsymbol{\theta})$. We have

$$\mathsf{mge}(\boldsymbol{\theta}, y^*) \geq \max_{j \in [m]} w_j |L_j(f(\boldsymbol{\theta}))|(y^* - f(\boldsymbol{\theta})).$$

We next discuss by cases.

1. **When** $L_g(f(\boldsymbol{\theta})) = 0$, there are at most $m - 1$ groups on the left of $f(\boldsymbol{\theta})$. By the definition of the `MEDIAN` mechanism, there are at least $\frac{n}{2}$ agents on the left of $f(\boldsymbol{\theta})$ and at most $\frac{n}{2}$ agents on the right of $f(\boldsymbol{\theta})$. Hence, we have $\max_{j \in [m]} |L_j(f(\boldsymbol{\theta}))| \geq \frac{n}{2(m-1)}$ and $|R_g(f(\boldsymbol{\theta}))| \leq \frac{n}{2}$. Therefore, we can bound the approximation ratio by

$$\begin{aligned}\rho &= \frac{\mathsf{mge}(\boldsymbol{\theta}, f(\boldsymbol{\theta}))}{\mathsf{mge}(\boldsymbol{\theta}, y^*)} \\ &\leq \frac{\mathsf{mge}(\boldsymbol{\theta}, y^*) + w_g \cdot (|R_g(f(\boldsymbol{\theta}))| - |L_g(f(\boldsymbol{\theta})|)(y^* - f(\boldsymbol{\theta}))}{\mathsf{mge}(\boldsymbol{\theta}, y^*)} \\ &\leq \frac{\mathsf{mge}(\boldsymbol{\theta}, y^*) + w_g \cdot |R_g(f(\boldsymbol{\theta}))|(y^* - f(\boldsymbol{\theta}))}{\mathsf{mge}(\boldsymbol{\theta}, y^*)} \\ &\leq 1 + \frac{w_g \cdot |R_g(f(\boldsymbol{\theta}))|(y^* - f(\boldsymbol{\theta}))}{\max_{j \in [m]} w_j \cdot |L_j(f(\boldsymbol{\theta}))|(y^* - f(\boldsymbol{\theta}))} \\ &\leq 1 + \frac{w_g \cdot \frac{n}{2}}{\max_{j \in [m]} w_j \cdot \frac{n}{2(m-1)}} \\ &\leq 1 + \frac{w_g \cdot (m-1)}{\max_{j \in [m]} w_j} \\ &\leq 1 + (m-1) \cdot \frac{w_{\max}}{w_{\min}}.\end{aligned}$$

2. **When** $L_g(f(\boldsymbol{\theta})) > 0$, we first rewrite the approximation ratio by

$$\begin{aligned}\rho =& \frac{\mathsf{mge}(\boldsymbol{\theta}, f(\boldsymbol{\theta}))}{\mathsf{mge}(\boldsymbol{\theta}, y^*)} \\ \leq& \frac{\mathsf{mge}(\boldsymbol{\theta}, y^*) + w_g(|R_g(f(\boldsymbol{\theta}))| - |L_g(f(\boldsymbol{\theta}))|)(y^* - f(\boldsymbol{\theta}))}{\mathsf{mge}(\boldsymbol{\theta}, y^*)} \\ \leq& 1 + \frac{w_g(|R_g(f(\boldsymbol{\theta}))| - |L_g(f(\boldsymbol{\theta}))|)(y^* - f(\boldsymbol{\theta}))}{\max_{j \in [m]} w_j |L_j(f(\boldsymbol{\theta}))|(y^* - f(\boldsymbol{\theta}))}.\end{aligned}$$

After that, we further consider two sub-cases by discussing whether the group G_g achieves $\max_{j \in [m]} |L_j(f(\boldsymbol{\theta}))|$.

   (a) **When** $\max_{j \in [m]} |L_j(f(\boldsymbol{\theta}))|$ **is achieved by group** g**.** By the definition of the `MEDIAN` mechanism, there are at least $\frac{n}{2}$ agents on the left of $f(\boldsymbol{\theta})$ and at most $\frac{n}{2}$ agents on the right of $f(\boldsymbol{\theta})$. Since there are at most m groups, by the pigeonhole principle, we have $|L_g(f(\boldsymbol{\theta}))| \geq \frac{\frac{n}{2}}{m}$. We have the approximation ratio

$$\begin{aligned}\rho &\leq 1 + \frac{w_g(|R_g(f(\boldsymbol{\theta}))| - |L_g(f(\boldsymbol{\theta}))|)(y^* - f(\boldsymbol{\theta}))}{w_g |L_j(f(\boldsymbol{\theta}))|(y^* - f(\boldsymbol{\theta}))} \\ &= 1 + \frac{\big(|R_g(f(\boldsymbol{\theta}))| - |L_g(f(\boldsymbol{\theta}))|\big)}{|L_g(f(\boldsymbol{\theta}))|} \\ &= \frac{|R_g(f(\boldsymbol{\theta}))|}{|L_g(f(\boldsymbol{\theta}))|} \leq \frac{\frac{n}{2}}{\frac{\frac{n}{2}}{m}} = m.\end{aligned}$$

(b) **When $\max_{j\in[m]} |L_j(f(\boldsymbol{\theta}))|$ is not achieved by group g.** By the pigeonhole principle, we derive that

$$\max_{j\in[m]} |L_j(f(\boldsymbol{\theta}))| \geq \frac{\frac{n}{2} - |L_g(f(\boldsymbol{\theta}))|}{m-1}.$$

With the inequality in hand, we upper-bound the approximation ratio by

$$\begin{aligned}
\rho &\leq 1 + \frac{w_g(|R_g(f(\boldsymbol{\theta}))| - |L_g(f(\boldsymbol{\theta}))|)(y^* - f(\boldsymbol{\theta}))}{\max_{j\in[m]} w_j|L_j(f(\boldsymbol{\theta}))|(y^* - f(\boldsymbol{\theta}))} \\
&\leq 1 + \frac{w_g\left(|R_g(f(\boldsymbol{\theta}))| - |L_g(f(\boldsymbol{\theta}))|\right)}{\max_{j\in[m]} w_j \frac{\frac{n}{2}-|L_g(f(\boldsymbol{\theta}))|}{m-1}} \\
&\leq 1 + \frac{w_{\max}}{w_{\min}} \cdot \frac{\frac{n}{2} - |L_g(f(\boldsymbol{\theta}))|}{\frac{\frac{n}{2}-|L_g(f(\boldsymbol{\theta}))|}{m-1}} \\
&\leq 1 + (m-1) \cdot \frac{w_{\max}}{w_{\min}}.
\end{aligned}$$

Lastly, we provide an example to show that our analysis for the approximation ratio of `MEDIAN` mechanism is tight. Consider an instance where $m-1$ agents from different groups $G_2, \ldots, G_m$ with weight $w_{\min}$ are located at 0 and the other $m-1$ agents in G_1 with weight $w_{\max}$ are located at 1. The `MEDIAN` mechanism locates the facility at 0 achieving an mge of $(m-1)\cdot w_{\max}$, while locating the facility at $\frac{(m-1)\cdot w_{\max}}{(m-1)\cdot w_{\max}+w_{\min}}$ can achieve the optimal mge of $\frac{(m-1)\cdot w_{\max}\cdot w_{\min}}{(m-1)\cdot w_{\max}+w_{\min}}$. Hence the approximation ratio of the `MEDIAN` mechanism is at least $1 + (m-1)\cdot \frac{w_{\max}}{w_{\min}}$, which completes the proof. □

We next prove that the `LEFTMOST` mechanism achieves an approximation ratio of $1 + (n-1)\cdot \frac{w_{\max}}{w_{\min}}$ for minimizing the mge objective when $E_j = w_j \cdot \sum_{i\in G_j} c(f(\boldsymbol{\theta}), x_i)$.

Proposition A.2. *The* `LEFTMOST` *mechanism has an approximation ratio of* $1+(n-1)\cdot \frac{w_{\max}}{w_{\min}}$ *for minimizing* mge *when* $E_j = w_j \cdot \sum_{i\in G_j} c(f(\boldsymbol{\theta}), x_i)$.

Proof. Consider any profile $\boldsymbol{\theta}$. Let $f(\boldsymbol{\theta})$ be the output of the `LEFTMOST` mechanism and y^* be the optimal facility location that minimizes mge objective. Without loss of generality, assume that $\mathsf{mge}(\boldsymbol{\theta}, f(\boldsymbol{\theta}))$ is achieved by group G_g. By definition of the `LEFTMOST` mechanism, we first have $f(\boldsymbol{\theta}) \leq y^*$. We use the analogous relocation approach as that in the proof of Theorem A.1 and focus on the instances after relocation. Note that $\mathsf{mge}(\boldsymbol{\theta}, f(\boldsymbol{\theta})) = w_g \sum_{i\in G_g} d(x_i, f(\boldsymbol{\theta}))$ and $\mathsf{mge}(\boldsymbol{\theta}, y^*) \geq w_g \sum_{i\in G_g} d(x_i, y^*)$. Similar to the proof of Theorem A.1, we have

$$\mathsf{mge}(\boldsymbol{\theta}, f(\boldsymbol{\theta})) - \mathsf{mge}(\boldsymbol{\theta}, y^*) \leq w_g(|R_g(f(\boldsymbol{\theta}))| - |L_g(f(\boldsymbol{\theta}))|)(y^* - f(\boldsymbol{\theta})),$$

and since there are $|L_j(f(\boldsymbol{\theta}))|$ agents in G_j at or on the left of $f(\boldsymbol{\theta})$, we have

$$\mathsf{mge}(\boldsymbol{\theta}, y^*) \geq \max_{j\in[m]} w_j|L_j(f(\boldsymbol{\theta}))|(y^* - f(\boldsymbol{\theta})).$$

So the approximation ratio is bounded by

$$\begin{aligned}
\rho =& \frac{\mathsf{mge}(\boldsymbol{\theta}, f(\boldsymbol{\theta}))}{\mathsf{mge}(\boldsymbol{\theta}, y^*)} \\
\leq& 1 + \frac{w_g(|R_g(f(\boldsymbol{\theta}))| - |L_g(f(\boldsymbol{\theta}))|)(y^* - f(\boldsymbol{\theta}))}{\max_j w_j|L_j(f(\boldsymbol{\theta}))|(y^* - f(\boldsymbol{\theta}))} \\
=& 1 + \frac{w_{\max}}{w_{\min}} \cdot \frac{|R_g(f(\boldsymbol{\theta}))| - |L_g(f(\boldsymbol{\theta}))|}{\max_j |L_j(f(\boldsymbol{\theta}))|}.
\end{aligned}$$

Notice that $\max_{j\in[m]} |L_j(y)| \geq 1$ and $|R_g(y)| - |L_g(f(\boldsymbol{\theta}))| \leq n-1$. The approximation ratio is further bounded by

$$\begin{aligned}
\rho &= 1 + \frac{w_{\max}}{w_{\min}} \cdot \frac{|R_g(f(\boldsymbol{\theta}))| - |L_g(f(\boldsymbol{\theta}))|}{\max_j |L_j(f(\boldsymbol{\theta}))|} \\
&\leq 1 + (n-1) \cdot \frac{w_{\max}}{w_{\min}}.
\end{aligned}$$

The following example shows that our analysis for the approximation ratio of the LEFTMOST mechanism is tight. Consider an instance with two groups where one agent in G_1 with $w_1 = w_{\min}$ at 0, and $n-1$ agents in G_2 with $w_2 = w_{\max}$ at 1. The optimal mge solution locates the facility at $\frac{(n-1)\cdot w_{\max}}{(n-1)\cdot w_{\max}+w_{\min}}$, achieving an mge of $\frac{(n-1)w_{\max}\cdot w_{\min}}{(n-1)w_{\max}+w_{\min}}$, and the LEFTMOST mechanism locates the facility at 0, achieving an mge of $(n-1)\cdot w_{\max}$. Hence, the approximate ratio of the LEFTMOST mechanism is at least $1+(n-1)\cdot\frac{w_{\max}}{w_{\min}}$. □

We next turn our attention to the MAJOR mechanism which was proposed by Zhou et al. [18]. Under the mge objective, they showed that when $w_j = 1$ (i.e., maximum total group cost, mtgc) or $w_j = \frac{1}{|G_j|}$ (i.e., maximum average group cost, magc), the MAJOR mechanism achieves an approximation ratio of 3 in both cases. However, we show that the MAJOR mechanism is not robust: when arbitrary group weights w_j are allowed under the mge objective, it no longer guarantees a 3-approximation.

Proposition A.3. *The* MAJOR *mechanism has an approximation ratio of* $1+2\cdot\frac{w_{\max}}{w_{\min}}$ *for minimizing* mge *objective when* $E_j = w_j\cdot\sum_{i\in G_j} c(f(\boldsymbol{\theta}), x_i)$.

Proof. Consider any arbitrary profile $\boldsymbol{\theta}$. Let $f(\boldsymbol{\theta})$ be the output of the MAJOR mechanism and y^* be the optimal location. Without loss of generality, assume that $f(\boldsymbol{\theta}) < y^*$ and $\mathsf{mge}(\boldsymbol{\theta}, f(\boldsymbol{\theta}))$ is achieved by group G_g. Recall Algorithm 3 and the group G_ℓ with largest size of agents. We first observe that there are at least $\frac{|G_\ell|}{2}$ agents in group G_ℓ on the left side of $f(\boldsymbol{\theta})$, which gives us

$$\mathsf{mge}(\boldsymbol{\theta}, y^*) \geq w_\ell\cdot\frac{|G_\ell|}{2}\cdot(y^* - f(\boldsymbol{\theta})).$$

Similar to the proof of Theorem A.1, we have

$$\mathsf{mge}(\boldsymbol{\theta}, f(\boldsymbol{\theta})) - \mathsf{mge}(\boldsymbol{\theta}, y^*) \leq w_g(|R_g(f(\boldsymbol{\theta}))| - |L_g(\boldsymbol{\theta})|)(y^* - f(\boldsymbol{\theta})).$$

Finally, the approximation ratio is bounded by

$$\begin{aligned}
\rho &= \frac{\mathsf{mge}(\boldsymbol{\theta}, f(\boldsymbol{\theta}))}{\mathsf{mge}(\boldsymbol{\theta}, y^*)} \\
&\leq 1 + \frac{w_g(|R_g(f(\boldsymbol{\theta}))| - |L_g(\boldsymbol{\theta})|)(y^* - f(\boldsymbol{\theta}))}{\mathsf{mge}(\boldsymbol{\theta}, y^*)} \\
&\leq 1 + \frac{w_g|G_g|(y^* - f(\boldsymbol{\theta}))}{\mathsf{mge}(\boldsymbol{\theta}, y^*)} \\
&\leq 1 + \frac{w_g\cdot|G_g|\cdot(y^* - f(\boldsymbol{\theta}))}{w_\ell\cdot\frac{|G_\ell|}{2}\cdot(y^* - f(\boldsymbol{\theta}))} \\
&\leq 1 + 2\cdot\frac{w_g}{w_\ell}\cdot\frac{|G_g|}{|G_\ell|} \\
&\leq 1 + 2\cdot\frac{w_{\max}}{w_{min}}.
\end{aligned}$$

When considering $w_j = 1$ or $w_j = \frac{1}{|G_j|}$, we have $w_{\max} = w_{\min}$ which implies $1+2\cdot\frac{w_{\max}}{w_{\min}}$ degenerates to an approximation ratio of 3. We provide the following example that shows our analysis for the approximation ratio of the MAJOR mechanism is tight. Consider an instance with one agent in group G_1 with $w_{\min}$ at 0, one agent in group G_1 at $\frac{2\cdot w_{\max}}{2\cdot w_{\max}+w_{\min}}$, and two agents in group G_2 with $w_{\max}$ at 1. The MAJOR mechanism locates the facility at 0, achieving the mge of $2\cdots w_{\max}$, however, locating the facility at $\frac{2w_{\max}}{2w_{\max}+w_{\min}}$ achieves the optimal mge value at $\frac{2w_{\max}\cdot w_{\min}}{2w_{\max}+w_{\min}}$. Hence, the approximation ratio of the MAJOR mechanism is at least $1+2\frac{w_{\max}}{w_{min}}$. □

A.2 Weighted Maximum Group Cost

In this subsection, we consider maximum group effect defined by $E_j = w_j\cdot\max_{i\in G_j} c(f(\theta), x_i)$. We first show that locating the facility at any agent location can achieve an approximation ratio of $1+\frac{w_{\max}}{w_{\min}}$.

Lemma A.4. *Given any agent profile* $\boldsymbol{\theta}$*, Locating the facility at any agent* i*'s location* x_i *achieves an approximation ratio of* $1+\frac{w_{\max}}{w_{\min}}$.

Proof. Given any profile $\boldsymbol{\theta}$, without loss of generality, assume that the facility is located at x_k, and $\mathsf{mge}(\boldsymbol{\theta}, x_k)$ is incurred by agent ℓ, i.e., $\mathsf{mge}(\boldsymbol{\theta}, x_k) = w_{g_\ell} \cdot d(x_\ell, x_k)$[3]. Let y^* be the optimal facility location. Without loss of generality, we assume that $x_k \le y^*$.

Case (1). If $x_\ell \le x_k \le y^*$, we have

$$\mathsf{mge}(\boldsymbol{\theta}, y^*) \ge w_{g_k} \cdot d(x_k, y^*) \ge w_{g_k} \cdot d(x_\ell, x_k).$$

So the approximation ratio can be upper-bounded by

$$\rho = \frac{\mathsf{mge}(\boldsymbol{\theta}, x_k)}{\mathsf{mge}(\boldsymbol{\theta}, y^*)} \le \frac{w_{g_\ell} \cdot d(x_\ell, x_k)}{w_{g_k} \cdot d(x_\ell, x_k)} \le \frac{w_{\max}}{w_{\min}}.$$

Case (2). If $x_k < x_\ell \le y^*$, we have

$$\mathsf{mge}(\boldsymbol{\theta}, y^*) \ge w_{g_\ell} \cdot d(x_\ell, y^*) \ge w_{g_\ell} \cdot d(x_\ell, x_k).$$

which implies an optimum.

Case (3). If $x_k < y^* < x_\ell$, from Theorem 3.5, we derive that

$$\begin{aligned}\mathsf{mge}(\boldsymbol{\theta}, y^*) &\ge \max_{k,\ell} \left\{ w_{g_k} \cdot d(x_k, y^*), w_{g_\ell} \cdot d(x_\ell, y^*) \right\} \\ &\ge \frac{w_{g_k} \cdot w_{g_\ell} \cdot d(x_\ell, x_k)}{w_{g_k} + w_{g_\ell}}.\end{aligned}$$

Consequently, we further bound the approximation ratio by

$$\rho = \frac{\mathsf{mge}(\boldsymbol{\theta}, x_k)}{\mathsf{mge}(\boldsymbol{\theta}, y^*)} \le \frac{w_{g_\ell} \cdot d(x_\ell, x_k)}{\frac{w_{g_k} \cdot w_{g_\ell} \cdot d(x_\ell, x_k)}{w_{g_k} + w_{g_\ell}}} \le \frac{w_{g_k} + w_{g_\ell}}{w_{g_k}} \le 1 + \frac{w_{\max}}{w_{\min}}.$$

□

Proposition A.5. *The* MEDIAN, LEFTMOST, *and* MAJOR *mechanisms have an approximation ratio of* $1 + \frac{w_{\max}}{w_{\min}}$ *for minimizing* mge *when* $E_j = w_j \cdot \max_{i \in G_j} c(f(\boldsymbol{\theta}), x_i)$.

Proof. By Theorem A.4, we immediately have that the MEDIAN, LEFTMOST, and MAJOR mechanisms have an approximation ratio of at most $1 + \frac{w_{\max}}{w_{\min}}$. To show the tightness, we then show that the approximation ratio is at least $1 + \frac{w_{\max}}{w_{\min}}$. Consider an example with one agent in group G_1 with $w_1 = w_{\min}$ at 0 and one agent in group G_2 with $w_2 = w_{\max}$ at 1. For any of these three mechanisms, locating the facility at 0 achieves the mge objective of $w_{\max}$. For the optimal location $\frac{w_{\max}}{w_{\max}+w_{\min}}$, it achieves the mge objective of $\frac{w_{\max} \cdot w_{\min}}{w_{\max}+w_{\min}}$. □

B Omitted Proofs from Section 3

B.1 Omitted Proof of Theorem 3.1

Proof. For $m = 1$ and equal group weight, mge degenerates to the classical social cost objective. In this case, BALANCED mechanism places the facility at $\min\{y : L_1(y) \ge R_1(y)\}$, which is the leftmost location y such that the number of agents to the left of y is at least as large as the number of agents to the right. This is precisely the definition of median point among the n agents[4]. For the maximum cost objective (i.e., $m = n$ and equal weights), the leftmost agent location will be selected by BALANCED mechanism since we have $L_1(x_1) = 1$ and $\max_{j \in [m]} R_j(x_1) = 1$. □

B.2 Omitted Proof of Theorem 3.5

Proof. Given any two-agent instance profile $\boldsymbol{\theta}$, we first observe that $y^* \in [x_1, x_2]$.

Case 1: $g_1 = g_2$**, i.e., two agents with the same group membership**. Given any facility location y, $\mathsf{mge}(\boldsymbol{\theta}, y) = w_1 \cdot \max\{|y - x_1|, |y - x_2|\}$. For $x_1 < x_2$, the function $\max\{|y - x_1|, |y - x_2|\}$ is piecewise

[3]Recall that w_{g_i} denotes the maximum weight among all he groups to which agent i belongs, i.e., $w_{g_i} = \max_{g \in g_i} w_g$.

[4]Breaking ties by selecting the left median when n is even.

linear and convex, minimized where $|y - x_1| = |y - x_2|$. By solving $y - x_1 = x_2 - y$, we have $y^* = \frac{x_1+x_2}{2}$. When $y^* = \frac{x_1+x_2}{2}$, the maximum group effect under y^* is

$$\mathsf{mge}(\boldsymbol{\theta}, y^*) = w_1 \cdot \frac{x_2 - x_1}{2}.$$

Case 2: $g_1 \neq g_2$, i.e., two agents with different group membership. Given any facility location y, $\mathsf{mge}(\boldsymbol{\theta}, y) = \max\{w_1|y - x_1|, w_2|y - x_2|\}$. For $x_1 < x_2$, minimize the convex function $\max\{w_1|y - x_1|, w_2|y - x_2|\}$. The minimum occurs where $w_1|y - x_1| = w_2|y - x_2|$. That is

$$w_1(y - x_1) = w_2(x_2 - y) \implies y^* = \frac{w_2x_2 + w_1x_1}{w_1 + w_2}.$$

At y^*:

$$w_1|y^* - x_1| = w_1\left(\frac{w_2x_2 + w_1x_1}{w_1 + w_2} - x_1\right) = \frac{w_1w_2(x_2 - x_1)}{w_1 + w_2},$$

and

$$w_2|y^* - x_2| = w_2\left(x_2 - \frac{w_2x_2 + w_1x_1}{w_1 + w_2}\right) = \frac{w_1w_2(x_2 - x_1)}{w_1 + w_2}.$$

Therefore, we have $\mathsf{mge}(\boldsymbol{\theta}, y^*) = \frac{w_1w_2(x_2-x_1)}{w_1+w_2}$. This completes the proof. □

C Omitted Proofs from Section 4

C.1 Omitted Proof of Theorem 4.3

Observe that the objective mge, where $E_j = w_j \cdot \sum_{i \in G_j} c(f(\boldsymbol{\theta}, x_i))$, strictly generalizes the classical social cost objective as a special case. We recall the characterization of deterministic strategyproof mechanisms for the case $k = 2$ due to Fotakis and Tzamos [7]. Specifically, when the number of the agents $n \geq 5$, a deterministic strategyproof mechanism f admits a bounded approximation ratio with respect to the social cost objective if and only if f is either the `ENDPOINT` mechanism or a `DICTATORIAL` mechanism. For the `ENDPOINT` mechanism, the approximation ratio is known to be $1 + (n-2) \cdot \frac{w_{\max}}{w_{\min}}$. We next analyze the `DICTATORIAL` mechanism. Let j be the dictator agent, so that one facility is placed at the dictator's reported location x_j. For the location of the other facility, the `DICTATORIAL` mechanism first considers the distance of x_j to the leftmost and to the rightmost location, $d_l = |\min \mathbf{x} - x_j|$ and $d_r = |\max \mathbf{x} - x_j|$, respectively. The second facility is then placed at $x_j - \max\{d_l, 2d_r\}$ if $d_l > d_r$, and $x_j + \max\{d_r, 2d_l\}$, otherwise. Consider the following instance with n agents. The dictator agent j is located at position 0. There are $n - 2$ agents located at position $\frac{1}{2}$, and one additional agent located at position 1. The agents are partitioned into two groups as follows: $G_1 = \{j\}$ with weight $w_1 = w_{\min}$, and $G_2 = \{1, 2, \cdots, j-1, j+1, \cdots, n\}$ with weight $w_2 = w_{\max}$. Under this instance, the `DICTATORIAL` mechanism outputs the facility locations $f(\boldsymbol{\theta}) = \{x_j = 0, 1\}$. This outcome coincides with that of the `ENDPOINT` mechanism on the same instance. Consequently, the approximation ratio attained by the `DICTATORIAL` mechanism in this case is also $1 + (n-2) \cdot \frac{w_{\max}}{w_{\min}}$.

C.2 Omitted Proof of Theorem 4.5

Observe that the objective mge, where $E_j = w_j \cdot \max_{i \in G_j} c(f(\boldsymbol{\theta}, x_i))$, strictly generalizes the classical maximum cost objective as a special case. We show that for $k \geq 2$ cases, every deterministic strategyproof mechanism with a bounded approximation ratio for minimizing the maximum cost have a bounded approximation ratio for minimizing the social cost. To see this, consider a deterministic strategyproof mechanism f with an unbounded approximation ratio for minimizing the social cost. It implies that there exists an instance that the optimal social cost is 0 and the social cost achieved by f is larger than 0. It further implies that the optimal maximum cost is 0 and the maximum cost achieved by f is larger than 0, under the same instance. Hence, it is sufficient to consider all deterministic strategyproof mechanisms with bounded approximation ratios for minimizing the social cost, which are characterized by Fotakis and Tzamos [7]. Specifically, when the number of the agents $n \geq 5$, a deterministic strategyproof mechanism f admits a bounded approximation ratio with respect to the social cost objective if and only if f is either the `ENDPOINT` mechanism or a

DICTATORIAL mechanism. Therefore, for the maximum cost objective, we only focus on the ENDPOINT mechanism and the DICTATORIAL mechanism. As we have shown in Theorem 4.4, the ENDPOINT mechanism achieves a tight approximation ratio of $1 + \frac{w_{\max}}{w_{\min}}$ for the wMGC objective. To complete the proof of the existence lower bound, it suffices to show that the approximation ratio of the DICTATORIAL mechanism is at least $1 + \frac{w_{\max}}{w_{\min}}$. To see this, consider j is the dictator, and the instance is as follows. $x_1 = x_2 = \cdots = x_{n-2} = 0, x_{n-1} = \frac{1}{2}$, and $x_j = 1$. The group structure is given by $G_1 = \{1\}$, and $G_2 = \{2, 3, \cdots, n-1, j\}$, with weights $w_1 = w_{\min}$ and $w_2 = w_{\max}$. By the definition of the DICTATORIAL mechanism, one facility is placed at 1 and the other facility is placed at 0. The outcome aligns with the ENDPOINT mechanism and the instance is, in fact, the exact instance which shows the tight bound of $1 + \frac{w_{\max}}{w_{\min}}$ for the mechanism, hence, it implies that the DICTATORIAL mechanism has an approximation ratio of at least $1 + \frac{w_{\max}}{w_{\min}}$. This completes the proof.